RPA Course Syllabus

Description: Robotic Process Automation curriculum from fundamentals through workflow design, UI automation, orchestration, and deployment.

Module 1: RPA Fundamentals

Module Description: Introduction to Robotic Process Automation, its benefits, lifecycle, and practical use cases.

Topic 1: Introduction to RPA

Topic Description:

Lesson 1: What is RPA?

Lesson Description: Understand software robots and rule-based business process automation.

Lesson 2: RPA Benefits and Use Cases

Lesson Description: Explore common automation opportunities in business operations.

Lesson 3: Attended vs Unattended Automation

Lesson Description: Compare automation types and execution models.

Topic 2: RPA Lifecycle and Tools

Topic Description:

Lesson 1: RPA Lifecycle

Lesson Description: Understand discovery, design, development, testing, and maintenance.

Lesson 2: RPA Platforms Overview

Lesson Description: Explore UiPath, Automation Anywhere, and Power Automate.

Lesson 3: Setting Up an RPA Environment

Lesson Description: Install an RPA development tool and create a first project.

Module 2: Workflow Design and Basic Automation

Module Description: Build simple automated workflows using activities, variables, and control flow.

Topic 1: Workflow Essentials

Topic Description:

Lesson 1: Creating a Workflow

Lesson Description: Create and run a basic automation project.

Lesson 2: Variables and Data Types

Lesson Description: Store and manipulate values within a workflow.

Lesson 3: Arguments and Reusable Components

Lesson Description: Pass data between reusable workflow components.

Topic 2: Control Flow

Topic Description:

Lesson 1: Conditions and Decisions

Lesson Description: Use If and Switch activities to control execution.

Lesson 2: Loops and Iteration

Lesson Description: Use For Each and While loops for repetitive tasks.

Lesson 3: Debugging Workflows

Lesson Description: Use breakpoints, logs, and step execution to troubleshoot.

Module 3: UI Automation and Data Handling

Module Description: Automate desktop and web interactions and process structured business data.

Topic 1: User Interface Automation

Topic Description:

Lesson 1: Selectors and UI Elements

Lesson Description: Identify and interact with desktop and web controls.

Lesson 2: Mouse and Keyboard Actions

Lesson Description: Automate clicks, typing, and navigation.

Lesson 3: Browser Automation

Lesson Description: Open pages, complete forms, and extract information.

Topic 2: Data Processing

Topic Description:

Lesson 1: Excel Automation

Lesson Description: Read, update, and write spreadsheet data.

Lesson 2: Data Tables and Filtering

Lesson Description: Organize, filter, and transform tabular records.

Lesson 3: PDF and Email Automation

Lesson Description: Extract document content and automate email workflows.

Module 4: Advanced RPA and Exception Handling

Module Description: Create reliable automations with error handling, orchestration, and reusable components.

Topic 1: Reliable Automation

Topic Description:

Lesson 1: Exception Handling

Lesson Description: Handle application and business exceptions with Try/Catch.

Lesson 2: Retries and Wait Strategies

Lesson Description: Improve reliability with timeouts, retries, and synchronization.

Lesson 3: Logging and Screenshots

Lesson Description: Capture execution details for diagnostics and auditing.

Topic 2: Orchestration and Reuse

Topic Description:

Lesson 1: Queues and Transactions

Lesson Description: Process work items using queue-based automation.

Lesson 2: Scheduling and Triggers

Lesson Description: Run unattended workflows on schedules or events.

Lesson 3: Assets and Credentials

Lesson Description: Use secure configuration and credential storage.

Module 5: RPA Projects, Testing and Deployment

Module Description: Develop, test, deploy, and monitor an end-to-end automation project.

Topic 1: Testing and Quality

Topic Description:

Lesson 1: Process Documentation

Lesson Description: Prepare process maps and solution design documents.

Lesson 2: Testing RPA Workflows

Lesson Description: Perform unit, integration, and user acceptance testing.

Lesson 3: Security and Best Practices

Lesson Description: Apply least privilege and protect sensitive business data.

Topic 2: Deployment and Capstone

Topic Description:

Lesson 1: Publishing and Deployment

Lesson Description: Package and deploy an automation to a runtime environment.

Lesson 2: Monitoring and Maintenance

Lesson Description: Track execution, failures, and automation performance.

Lesson 3: End-to-End RPA Project

Lesson Description: Build a practical invoice, report, or employee onboarding automation.